Correlational Analysis of Industry Payments and GLP-1 Prescribing Among SRHS-Affiliated Providers

June 2023 – December 2025

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Abstract

This analysis tests whether CMS Open Payments (“Sunshine Act”) industry payments to SRHS-affiliated providers are associated with their prescribing of GLP-1 receptor agonists. Both data sources were truncated to a single common window, June 1, 2023 – December 31, 2025, so that every comparison below uses the same time period. Within this window: 449 providers wrote 156,687 GLP-1 prescriptions, and 272 providers received 3,447 SRHS-flagged payments; 163 providers appear in both. Paid providers prescribe GLP-1s at nearly twice the rate of unpaid providers (27.7 vs. 17.6 per month), and payment volume correlates moderately with prescribing volume (Spearman $\rho = 0.41–0.53$). A manufacturer-specific brand-loyalty test - the sharpest available test of an actual influence mechanism - is weak and points in opposite directions for the two manufacturers with material payment volume (Novo Nordisk $\rho = –0.17$; Lilly $\rho = +0.21$). A before/after comparison around each provider's first payment shows a 53% increase in prescribing rate, but this pattern is not payment-specific: the same health-system-wide GLP-1 volume grew 60% over the window regardless of any individual payment, and an identical before/after test using an arbitrary date unrelated to any payment produces a statistically similar result. Taken together, the evidence supports a real association between payments and prescribing volume but not a causal claim that payments drive that volume.

1. Introduction

The Physician Payments Sunshine Act requires manufacturers to publicly report payments to covered recipients (physicians and teaching hospitals). A recurring question is whether such payments are associated with, or responsible for, prescribing behavior. This analysis examines that question for the GLP-1 drug class (Ozempic, Wegovy, Rybelsus, Mounjaro, Zepbound, Trulicity, Victoza, Saxenda, Byetta, Bydureon, Adlyxin, and generic/compounded equivalents) among SRHS-affiliated providers.

2. Data and Methods

2.1 Data sources and common window

Prescriptions are drawn from the SRHS ePrescribe “Rx Loopback” extract (dbo.fact_prescriptions), filtered to GLP-1 drug names, with all patient-identifying columns removed prior to analysis. Payments are drawn from CMS Open Payments General Payments data (dbo.facts_sunshine_act_data_all), filtered to SRHS_Provider = ‘Y’ and GLP-1 product names.

The two source extracts cover different native date ranges (prescriptions: 2023–2026; payments: 2018–2025). To make every comparison in this report apples-to-apples, both datasets were restricted to the single overlapping window June 1, 2023 through December 31, 2025 before any statistic below was computed. This drops 30,432 prescription rows (16%) and 3,262 payment rows (49%) that fall outside the common window.

2.2 Cohort (within window)

Quantity	Value
Prescriptions in window (of 187,119 total)	156,687
Payment records in window (of 6,709 total)	3,447
Unique prescribing NPIs in window	449
Unique paid NPIs in window	272
NPIs present in both (exact match)	163
Providers with ≥ 5 GLP-1 Rx (brand analysis)	300
Providers eligible for before/after comparison	86

NPI join: DoctorNPI (prescriptions) and Covered_Recipient_NPI (payments), both 10-digit CMS NPIs, exact string match.

2.3 Manufacturer / brand classification

Each prescription was assigned a manufacturer by substring match of the drug name

- Ozempic/Rybelsus/Wegovy/Victoza/Saxenda → Novo Nordisk
- Mounjaro/Zepbound/Trulicity → Lilly
- Byetta/Bydureon → AstraZeneca; Adlyxin → Sanofi

falling back to generic molecule name for compounded/generic-labeled entries. Payment rows were assigned the manufacturer listed in the payment record, collapsed to the same labels.

2.4 Statistical methods

Pearson r and Spearman ρ were used to measure association strength; Spearman is reported alongside Pearson because payment dollars and prescription counts are both heavily right-skewed. All confidence intervals are nonparametric bootstrap 95% percentile intervals (resampling with replacement); iteration counts and seeds are noted with each result. Provider prescribing rate is expressed as rx_per_month = total Rx ÷ (observation span in days ÷ 30.44), so providers observed for different lengths of time are compared on a common footing.

Before/after comparison: for each provider whose first in-window payment had ≥ 45 days of prescribing data on both sides, monthly prescribing rate was compared before vs. after that date. As a check on whether this pattern is payment-specific, the identical comparison was repeated using the midpoint of each provider's own observation window in place of the payment date - a date with no relationship to any actual payment. Full results of that check are in Appendix A; the main text reports only the summary needed to interpret the before/after finding correctly.

3. Results

3.1 Paid vs. unpaid providers

Group	n	Mean rx/mo	Median rx/mo	SD	Min	Max
Unpaid	286	17.63	8.04	19.28	0.09	121.76
Paid (≥ 1 payment)	163	27.74	26.60	21.12	0.09	105.89

Difference in means (paid − unpaid) = 10.10 rx/month, bootstrap 95% CI [6.07, 14.07] (5,000 resamples per group, seed = 1)

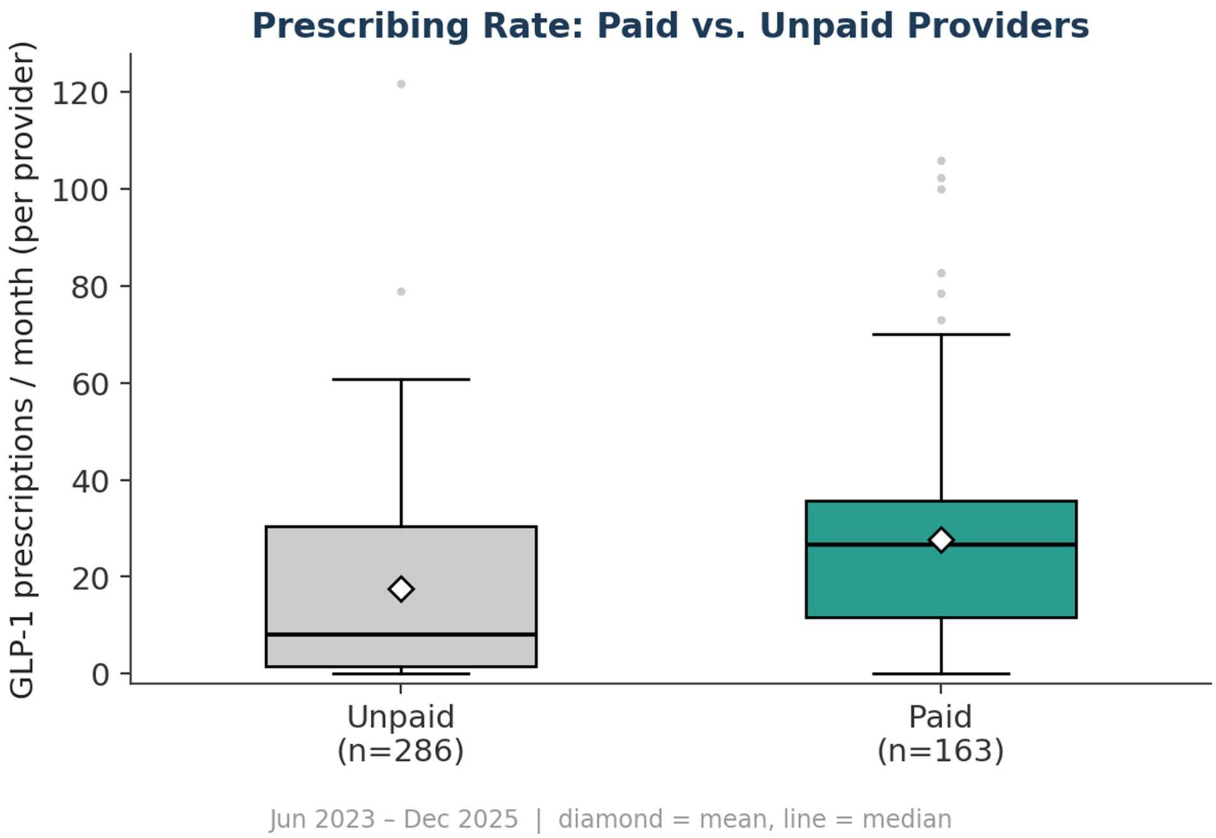

Figure 1. Prescribing rate distribution, paid vs. unpaid providers (Jun 2023–Dec 2025).

3.2 Payment volume vs. prescribing volume (paid providers, n=163)

Predictor	Outcome	Pearson r	95% CI	Spearman ρ
Total payment $	Total Rx count	0.457	[0.339, 0.574]	0.528
Total payment $	Rx per month	0.394	[0.264, 0.517]	0.410
Number of payments	Total Rx count	0.453	[0.336, 0.570]	0.534
Number of payments	Rx per month	0.387	[0.258, 0.508]	0.409

Bootstrap: 5,000 resamples, seed = 42, percentile method.

3.3 Brand-specific loyalty (providers with ≥ 5 GLP-1 Rx, n=300)

Tests whether being paid by a specific manufacturer associates with a higher share of that manufacturer's products in the provider's own prescribing mix - the most direct test available of a payment-specific influence mechanism, as opposed to general prescribing volume.

Manufacturer	$ vs. brand share - Pearson r	95% CI	Spearman ρ
Novo Nordisk	-0.113	[-0.206, -0.024]	-0.172
Lilly	0.168	[0.089, 0.247]	0.209

Bootstrap: 3,000 resamples, seed = 7, percentile method.

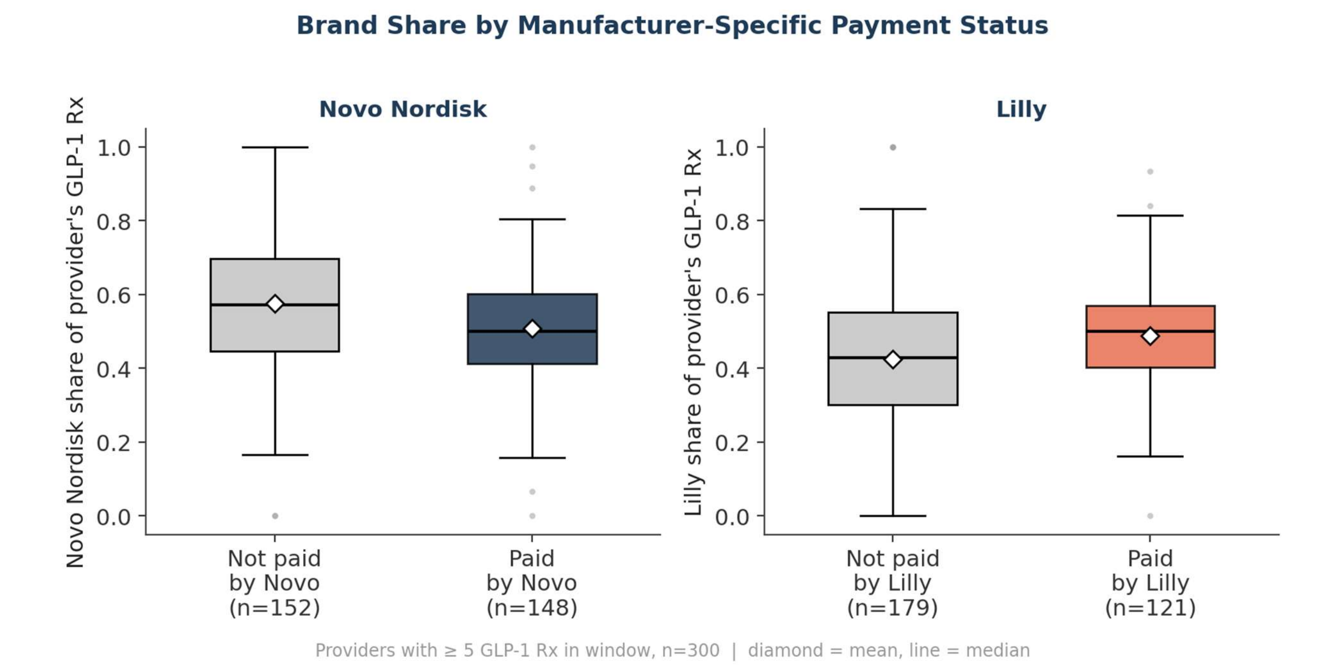

Figure 2. Brand share of GLP-1 prescribing, split by whether the provider was paid by that manufacturer. Novo Nordisk shows a small negative association; Lilly shows a small positive one - opposite directions.

3.4 Time trend: prescribing volume vs. payment volume

Across the full window, GLP-1 prescribing grew from 3,986 prescriptions in June 2023 to 6,383 in December 2025 (+60.2%), a steady system-wide climb. Monthly payment dollars, by contrast, fluctuate without a matching trend and fall sharply in the final two months of the window (November 2025: $178; December 2025: $154) while prescribing volume stays near its highest point of the series (5,328 and 6,383, respectively).

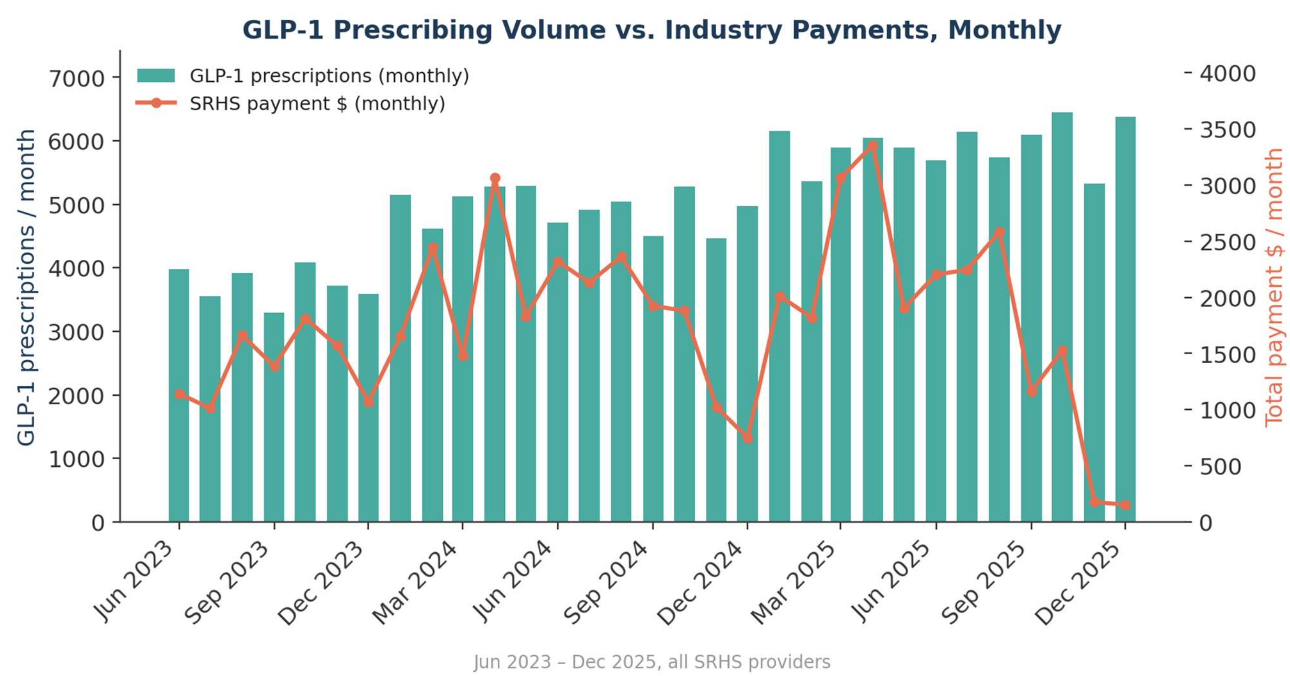

Figure 3. Monthly GLP-1 prescribing volume (bars, left axis) against monthly SRHS payment dollars (line, right axis), Jun 2023–Dec 2025.

3.5 Before vs. after a provider's first payment (n=86)

Metric	Before	After
Mean rx/month	17.52	26.84
Median rx/month	16.32	25.34
SD	12.37	18.32

Mean difference (after − before) = 9.32 rx/month, bootstrap 95% CI [6.96, 11.77] (5,000 resamples, seed = 11). 74 of 86 providers (86.0%) increased; relative change = +53.2%.

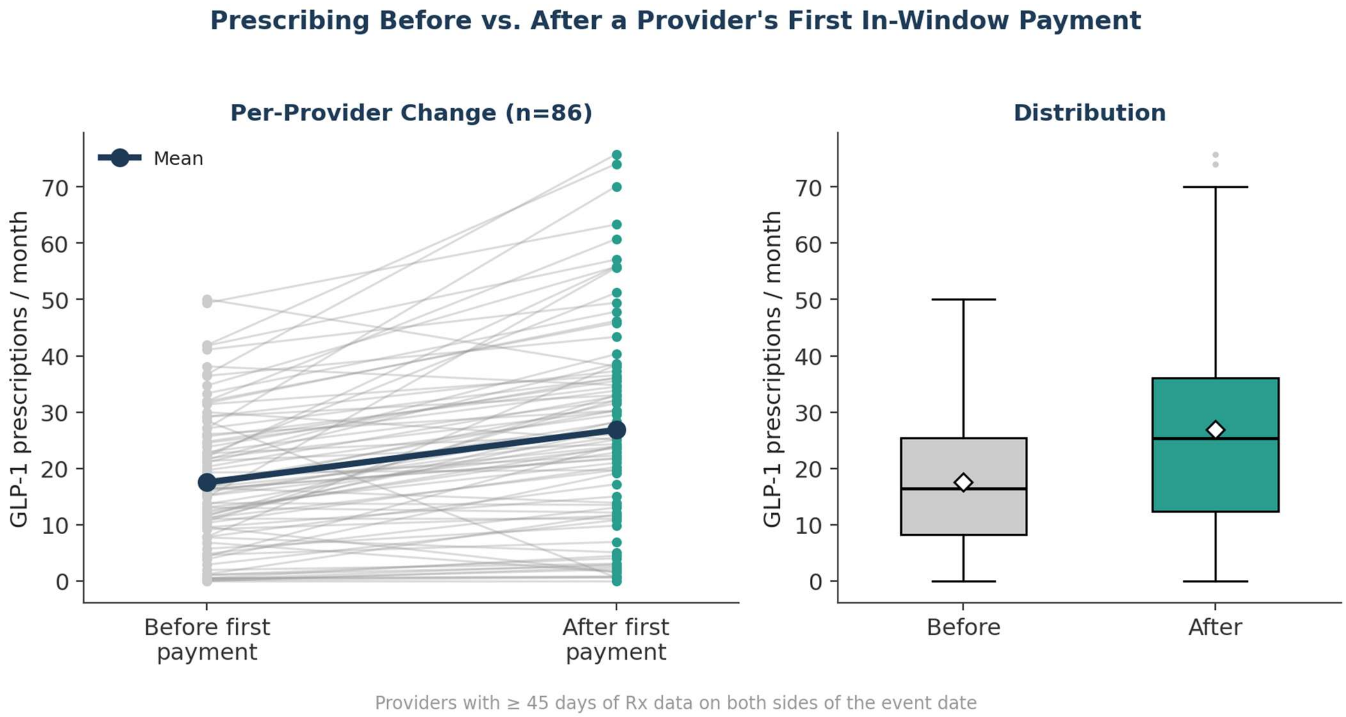

Figure 4. Per-provider prescribing rate before and after each provider's first in-window payment.

This looks like a payment effect, but Section 3.4 already shows why it should be read cautiously: system-wide GLP-1 volume grew 60% over the same window for reasons unrelated to any individual payment. Appendix A repeats this exact comparison using a date unrelated to any payment and finds a statistically similar increase, confirming that most of the pattern above reflects the broader time trend rather than the payment itself.

4. Discussion

The clearest, most defensible finding is the volume association in Sections 3.1–3.2: paid providers prescribe more GLP-1s, and payment intensity tracks prescribing intensity. This is real, but it is also the result least able to distinguish influence from targeting - manufacturers may simply direct outreach (see 4.1: nearly all of it is meals) toward providers who already prescribe heavily.

The brand-loyalty test in Section 3.3 is the sharpest available test of an actual payment-driven mechanism, and it does not hold up consistently: the direction of the association flips between the two manufacturers with meaningful payment volume in this data.

The time-based comparison in Sections 3.4–3.5 is the one most likely to be misread as causal, so it is presented with its own trend chart and, in Appendix A, a control test. Once the system-wide 60% growth in GLP-1 prescribing is visible on the same chart as payment dollars (Figure 3), the before/after jump in Figure 4 is much less surprising - nearly any two points in this window would show growth.

4.1 Limitations

• Within the window, payments are almost entirely Food and Beverage - small, frequent, low-dollar-value transfers, not consulting or speaking fees - so effect sizes should be read in that context.

• Reverse causation / targeting is not distinguishable from influence in this data: manufacturers may direct attention toward providers who are already high-volume prescribers.

• Confounding factors such as market adoption, extraneous exposure to advertisements not recorded in CMS Open Payment Data, and patient inquiries, not herein measured, can also play into increases in GLP-1 prescriptions during this period..

• No adjustment for specialty, panel size, years in practice, or patient case-mix - plausible confounders of both payment receipt and prescribing volume.

• The join between datasets is at the provider (NPI) level only; there is no encounter-level link between a specific payment and a specific prescription.

5. Conclusion

Within a common June 2023–December 2025 window, there is a real, moderate association between industry payments and GLP-1 prescribing volume among SRHS-affiliated providers. The two tests best suited to separate correlation from causation - manufacturer-specific brand loyalty and a controlled before/after comparison - do not support a causal payments-drive-prescribing interpretation. The apparent before/after increase tracks a system-wide 60% growth in GLP-1 prescribing over the same period; see Appendix A for the control test confirming this.

Appendix A: Placebo Control for the Before/After Comparison

To test whether the before/after result in Section 3.5 is specific to the payment date, the identical comparison was rerun for the same 86 providers using the midpoint of each provider's own observation window as the "event" - a date with no relationship to any actual payment.

Metric	**Real event (first payment)**	**Placebo (window midpoint)**
Mean difference (after − before)	9.32 rx/mo	7.45 rx/mo
Providers with an increase	74 / 86 (86.0%)	72 / 86 (83.7%)

The placebo reproduces almost the same result using a date unrelated to any payment, which is the basis for the caution urged in Sections 3.5 and 4.